

CubeSat Communications Board

70 cm BPSK TX / 2m AFSK RX

Project: CentiSat CubeSat -- Senior Capstone Design

Board: Communications Subsystem, Single PCB

Designer: Nick Grabbs / KI5YNG

Revsion: 0.5

Date: 05/20/2026

Repoitory: centisat/hardware/comms

Acronym Legend

* EPS - Electrical Power System

Key Design Notes

1. Power from EPS via CSKB bus – no on-board regulators
2. BPSK modulation at 145.67 MHz (before tripler)– 74LVC1G86 fails at 437 MHz, see overview.md timing analysis
3. Odd-order frequency multiplication preserves BPSK phase states ($3 \times 180^\circ = 540^\circ \equiv 180^\circ \bmod 360^\circ$)
4. Tripler bias 47k Ω – empirically optimized (-7.4 dBm @ 437 MHz)
5. All RF filter caps must be C0G/NP0 (not X7R)
6. I2C addresses: Si5351A = 0x60, EPS LTC4162 = 0x68 (no conflict)
7. Flat schematic design – nets connect by name across sheets
8. Prototype uses Pico module; bare RP2040 deferred to v2

Revision History

Rev	Date	Author	Description
0.1	2026-04-10	NG	Initial schematic– all sheets placed, basic connectivity

0.2 | 2026-04-13 | NG | Added Connectors sheet, stack bus, removed on-board regulators, added overview sheet |

0.3 | 2026-04-15 | NG | Added COMMS_IRQ on GP3 and SPI slave wiring on GP1619; initial Pumpkin pin-number lock |

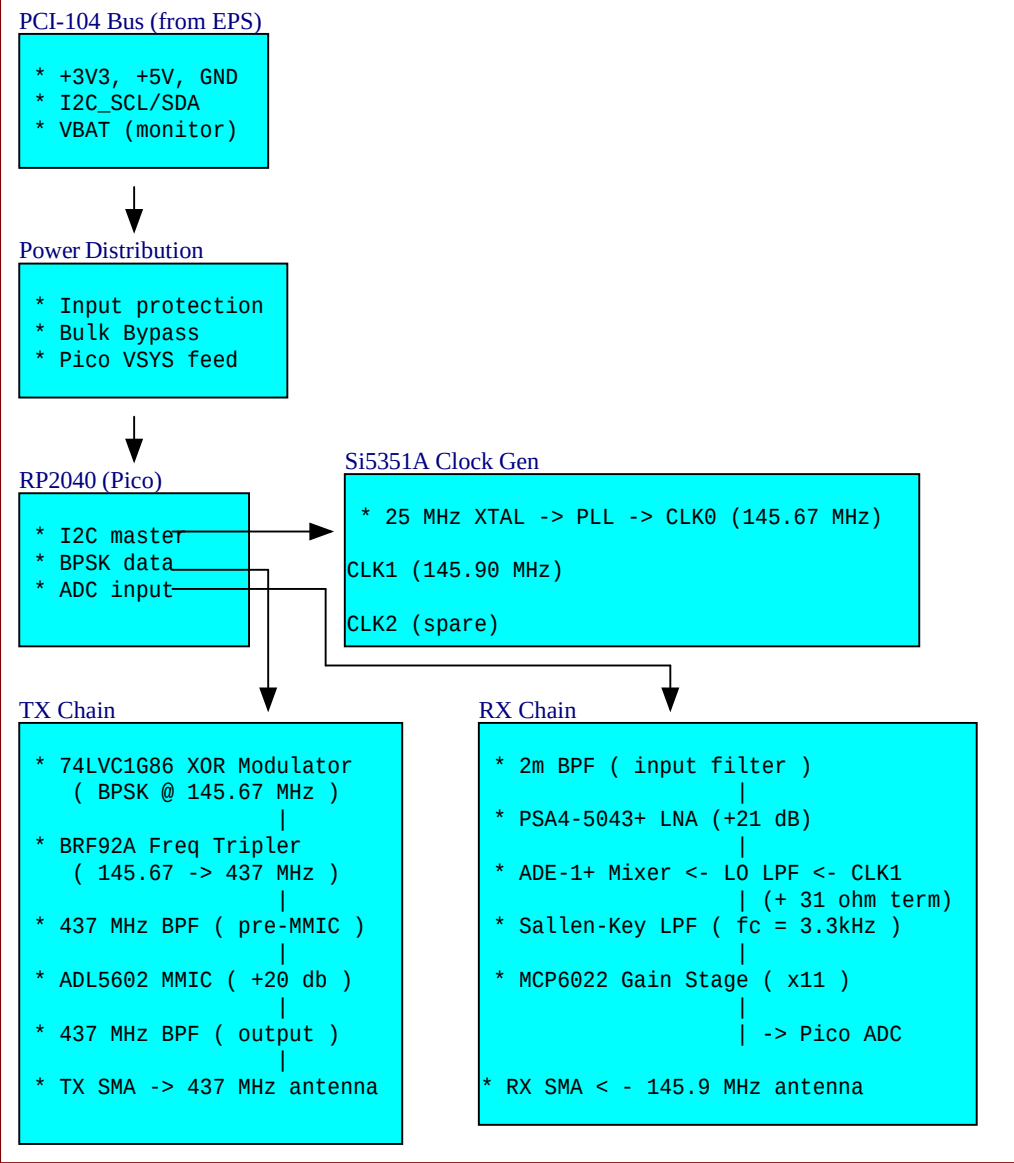
0.4 | 2026-04-15 | NG | Rebuilt connectors against Pumpkin datasheet Rev. E: bus renamed PC/104 → CubeSat Kit Bus (CSKB); pin numbers remapped H10 → H1/H2; connector family corrected to 2× Samtec ESQ-126-39-G-D stackthrough (2×26, 52 pins each); see `system/interfaces/cskb\_pinmap.md` |

0.5 | 2026-05-19 | NG | Sheet 4 RX front-end rebuilt: SA612 active mixer (EOL, sourcing failed) replaced with ADE-1+ passive diode mixer + PSA4-5043+ MMIC LNA + 5-pole LC LPF on the LO line + 51 Ω IF termination. New refdes: U10 (LNA), L12 (Vd choke), L13/L14 + C57/C58/C59 (LO LPF), R19 (IF term), C53/C54/C55/C56/C60 (DC blocks + bypass). Removed: U3 became ADE-1+ (was SA612), and C21/C22/C26/C27 + R7 (SA612-specific support parts) deleted. +5V budget grew from ~80 mA to ~140 mA. Rationale in `rx\_mixer\_trade\_study.md`; bench verification of LO drive level in `bringup/lo\_drive\_verification.md` and `bringup/si5351a\_breakout\_bench\_test.md`. |

Sheet Index

Sheet	Title	Contents
1	Overview	This sheet – block diagram, design notes
2	Clock Gen	Si5351A, 25 MHz crystal, I2C, clock outputs
3	TX Chain	XOR modulator, tripler, BPF, ADL5602 MMIC
4	RX Chain	PSA4 LNA, LO LPF, ADE-1+ mixer, IF term, Sallen-Key LPF, MCP6022 gain
5	Digital Control	RP2040 (Pico module), GPIOs, USB
6	Power	Distribution, protection, bypass
7	Connectors	CSKB H1 + H2 bus, SMA ports, test points

System Block Diagram



Power Rail Summary

Rail	Source	Voltage	Consumers
+3V3	EPS via CSKB (H2.27/28)	3.3V ±2%	RP2040, Si5351A, 74LVC1G86, MCP6022
+5V	EPS via CSKB (H2.25/26)	5.0V ±2%	PSA4-5043+ LNA, ADL5602 MMIC, tripler bias, Pico VSYS
GND	EPS via CSKB (H2.29/30/32, H2.31 AGND)	0V	All
VBAT	EPS via CSKB (H2.45/46)	6.0-8.4V	Monitor only (not used for power)

Estimated current budget:
+3V3: ~150 mA (RP2040 ~50mA, Si5351A ~30mA, logic ~10mA, op-amp ~10mA)
+5V: ~140 mA (PSA4 LNA ~60mA, ADL5602 ~60mA, tripler 5mA, Pico VSYS ~10mA, margin ~5mA)
Total: <300 mA – well within EPS TPS62933F capacity (3A per rail). Note: PSA4 LNA bias can be GPIO-gated for RX duty-cycling.

Net Name Convention

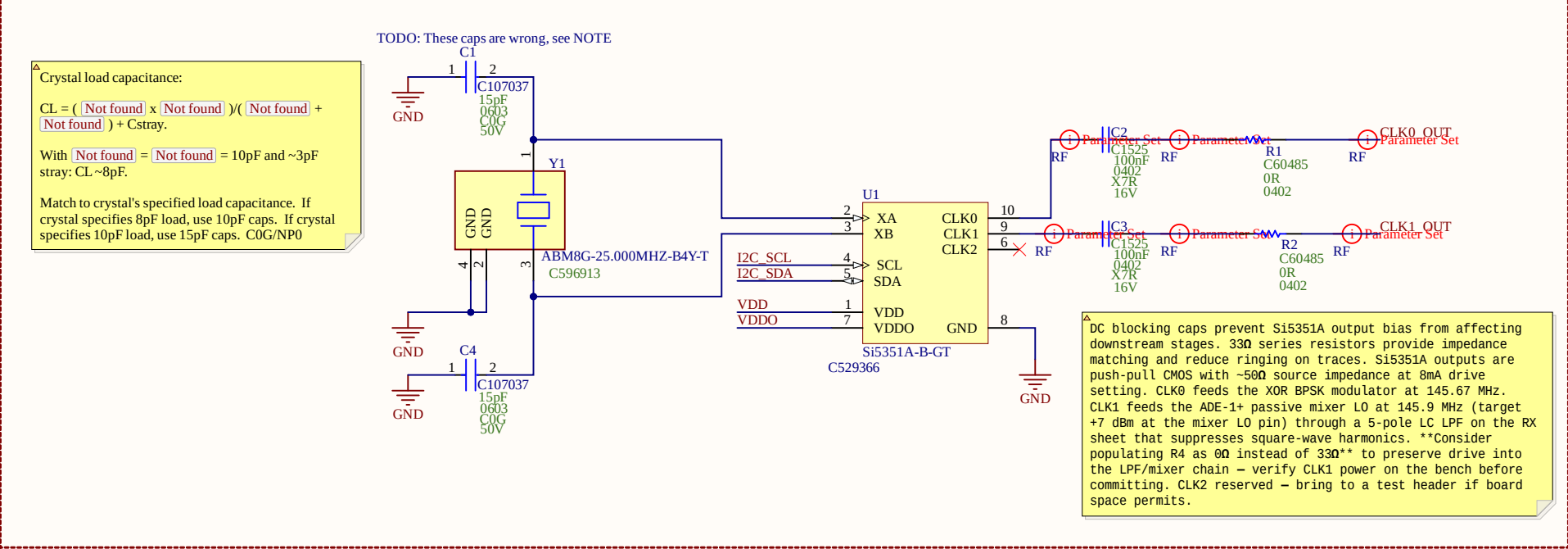
Net	Type	Description
+3V3, +5V, GND, VBAT	Power port	Supply rails (global)
I2C_SDA, I2C_SCL	Net label	I2C bus (Si5351A + IHU bus)
CLK0_OUT, CLK1_OUT	Net label	Si5351A clock outputs
BPSK_DATA	Net label	RP2040 → XOR modulator
TX_OUT, RX_IN	Net label	RF paths to SMA connectors
RX_BASEBAND	Net label	MCP6022 output → Pico ADC

CubeSat Comms -- Overview		
Size A3	Number	Revision A
Date: 8/15/2026	Sheet 1 of 7	
File: Overview.SchDoc	Drawn By: Nick Grabbs - KI5YNG	

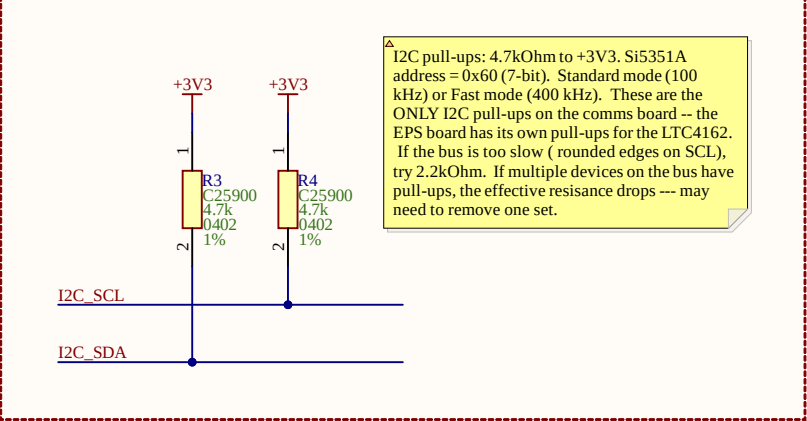
CLOCK GENERATOR

CLOCK GENERATOR -- Si5351A (I2C address 0x60). Generates three independent clock outputs from a single 25 MHz crystal. CLK0 = 145.67 MHz (TX carrier, tripled to 437 MHz). CLK1 = 145.9 MHz (RX local oscillator for ADE-1+ passive mixer; see RX sheet). CLK2 = spare. All outputs driven at 8 mA (max) into 500ohm loads. Controlled by RP2040 over I2C.

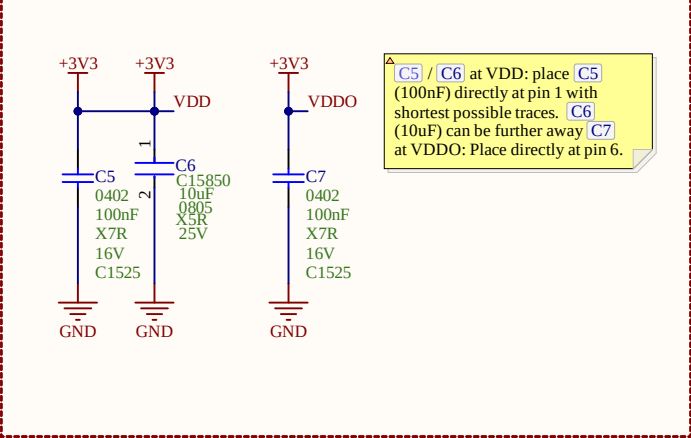
Si5351A IC



I2C Pull-Up Resistors



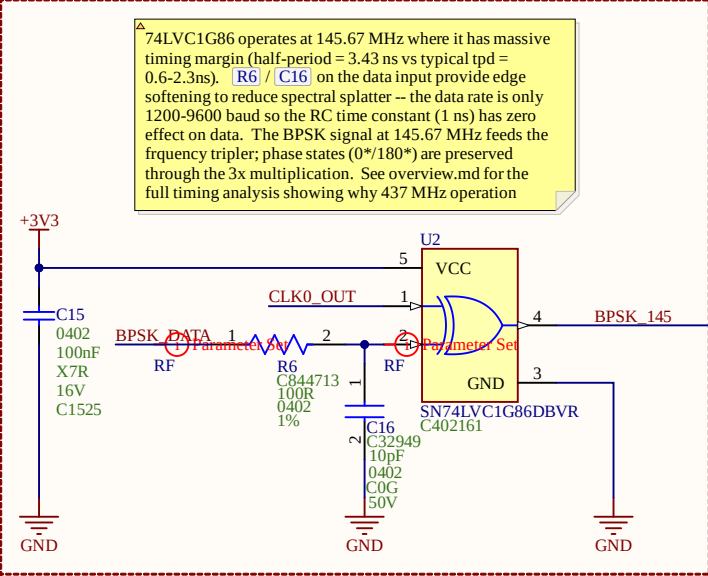
Bypass Capacitors



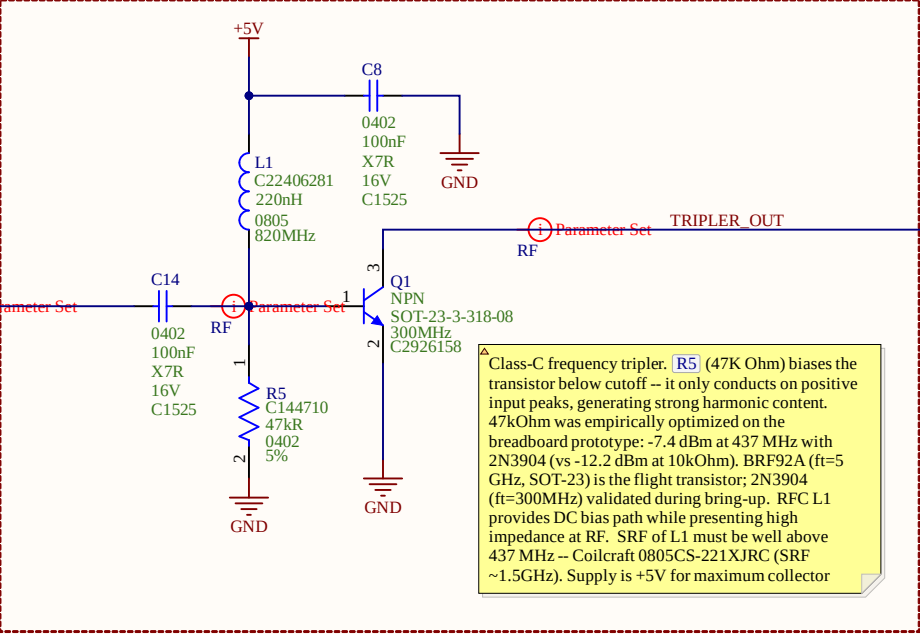
TX CHAIN

TX CHAIN - 437 MHz BPSK downlink transmitter. The carrier from the Si5351A (145.67 MHz) is BPSK-modulated by the 74LVC1G86 XOR gate, then frequency-tripled to 437 MHz by a class-C biased BFR92A transistor. A 3-pole LC BPF selects the 3rd harmonic and rejects the fundamental and 2nd harmonic. The ADL5602 MMIC provides ~20 dB gain to bring the signal to ~+10 dBm output. BPSK phase states are preserved through odd-order multiplication: $3 \times 180^\circ = 540^\circ = 180^\circ$

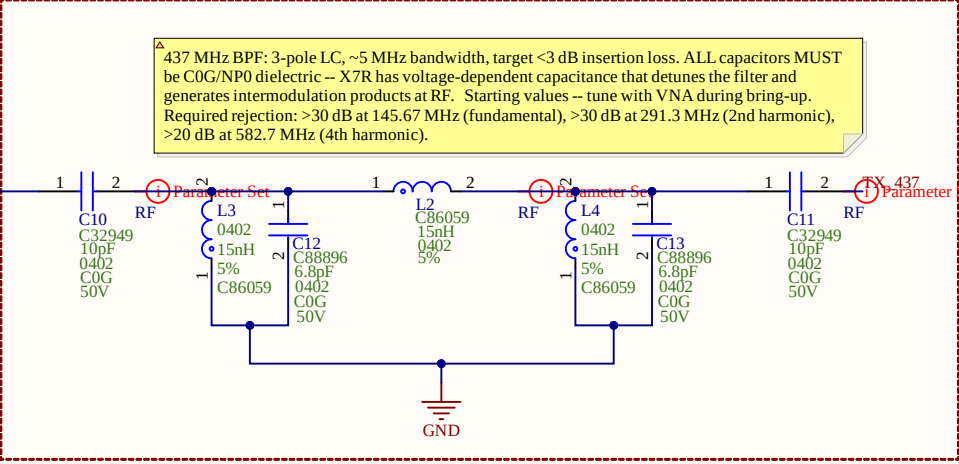
BPSK Modulator - 74LVC1G86 (SOT-23-5)



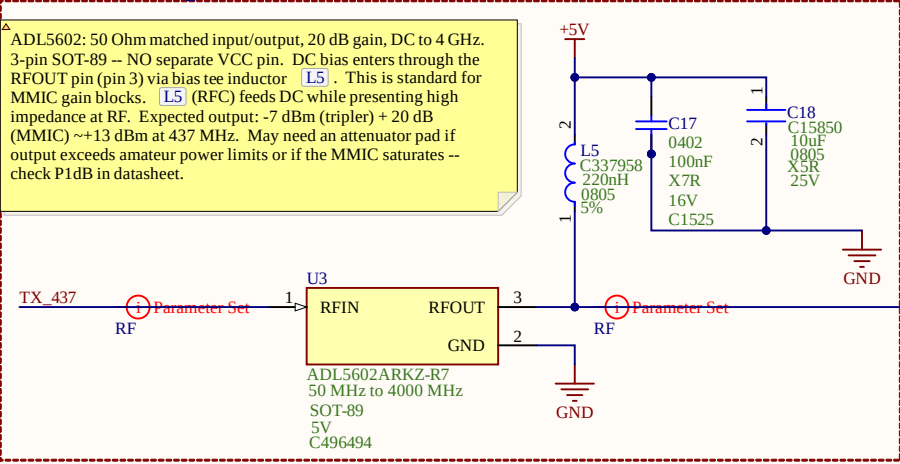
Frequency Tripler - BRF92A (SOT-23)



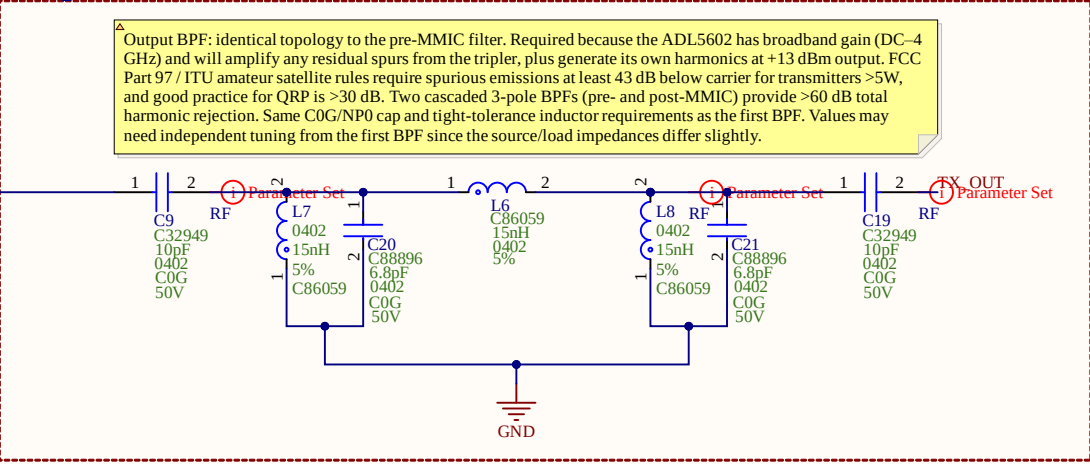
437 MHz Band-Pass Filter



MMIC Amplifier



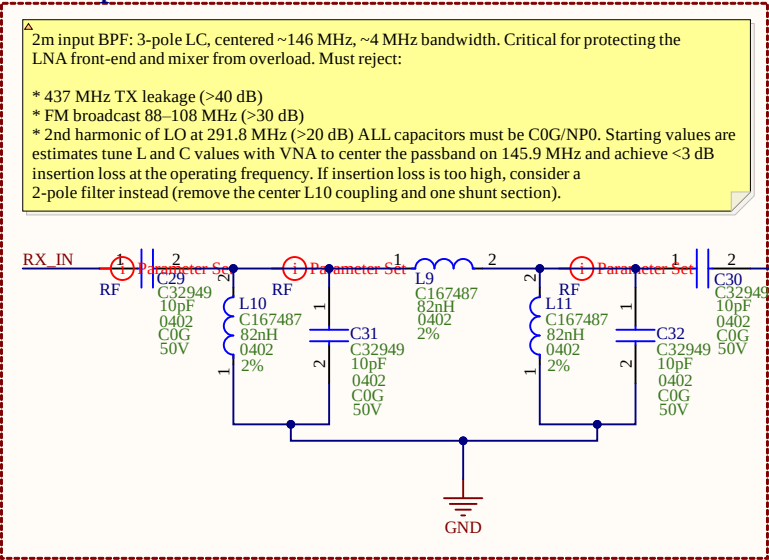
Output Band-Pass Filter



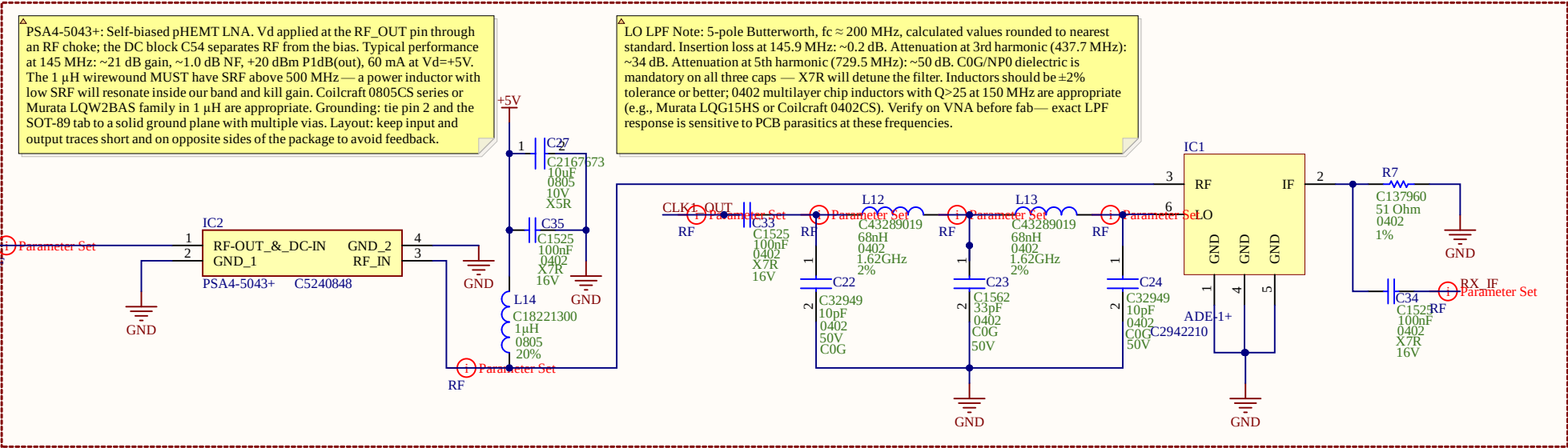
RX CHAIN

RX CHAIN – 145.9 MHz FM AFSK uplink receiver. A 2m band-pass filter at the input rejects out-of-band signals before the LNA. A PSA4-5043+ MMIC LNA (~21 dB gain, ~1 dB NF) sets the system noise figure and recovers the conversion loss of the downstream passive mixer. The ADE-1+ double-balanced diode mixer downconverts the 2m signal to baseband using the Si5351A CLK1 as LO; an LC low-pass on the LO line suppresses square-wave harmonics that would otherwise create spurious responses. A 51 Ω resistor terminates the mixer IF port. A 2nd-order Sallen-Key LPF (fc ≈ 3.3 kHz) filters the baseband, passing the 1200/2200 Hz AFSK tones while rejecting mixer products and noise. The MCP6022 U7B stage provides ~20 dB gain before the RP2040 ADC. Single-supply operation with 1.65V virtual ground (mid-supply bias). See rx_mixer_trade_study.md for the SA612 → ADE-1+ swap rationale.

2m Input Band-Pass Filter



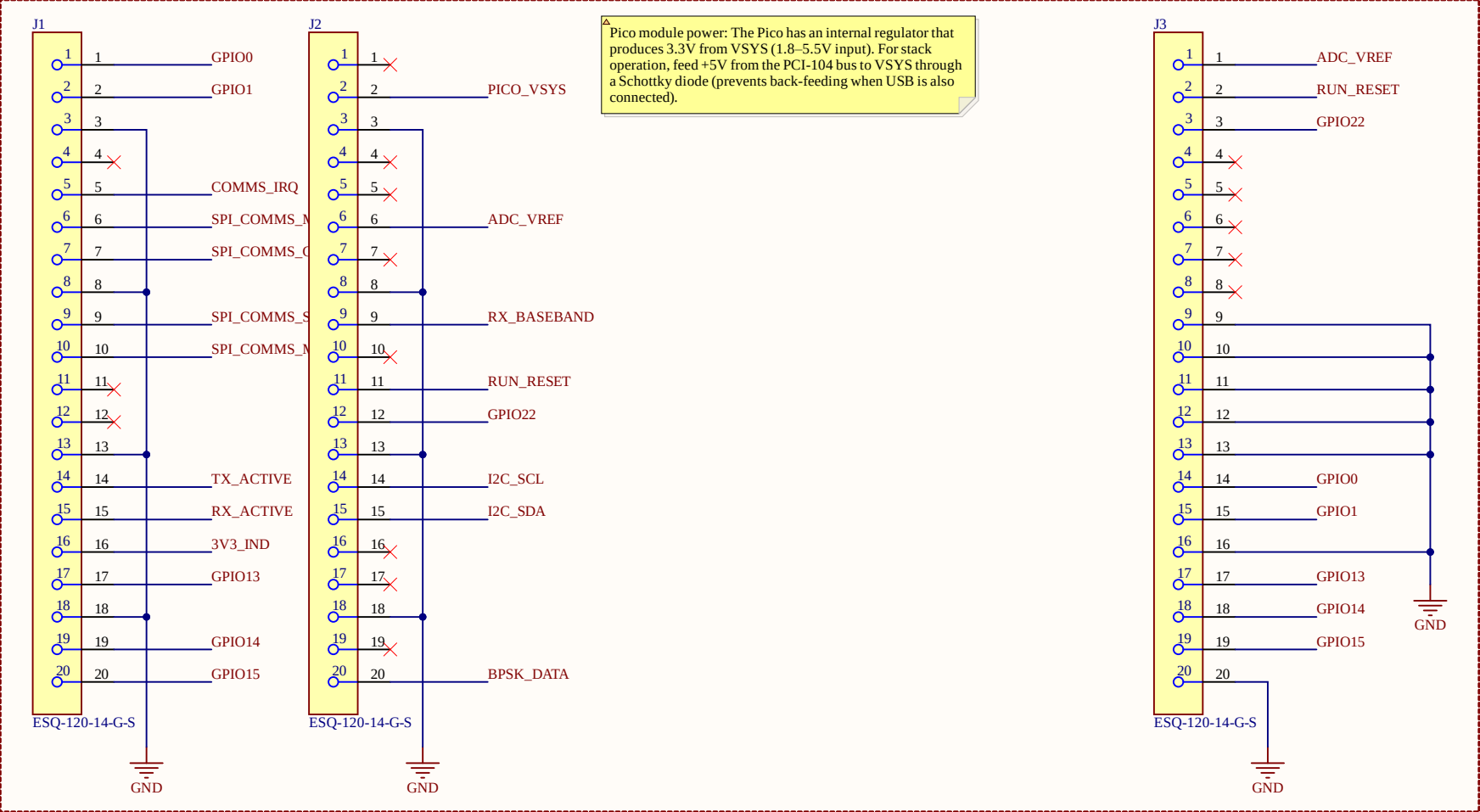
LNA -> LO LPF -> ADE-1+ Mixer -> IF Termination



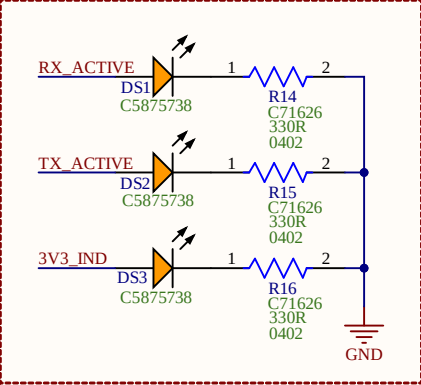
DIGITAL CONTROL

DIGITAL CONTROL — Raspberry Pi Pico module (RP2040-based). Handles BPSK/DBPSK encoding for TX, AFSK demodulation for RX, Si5351A configuration via I2C, and command processing. Using Pico module for prototype — all RP2040 support circuitry (flash, regulator, crystal) is on-module. Bare RP2040 redesign deferred to v2 flight board after RF chain validation.

Pico Module



Status LED's

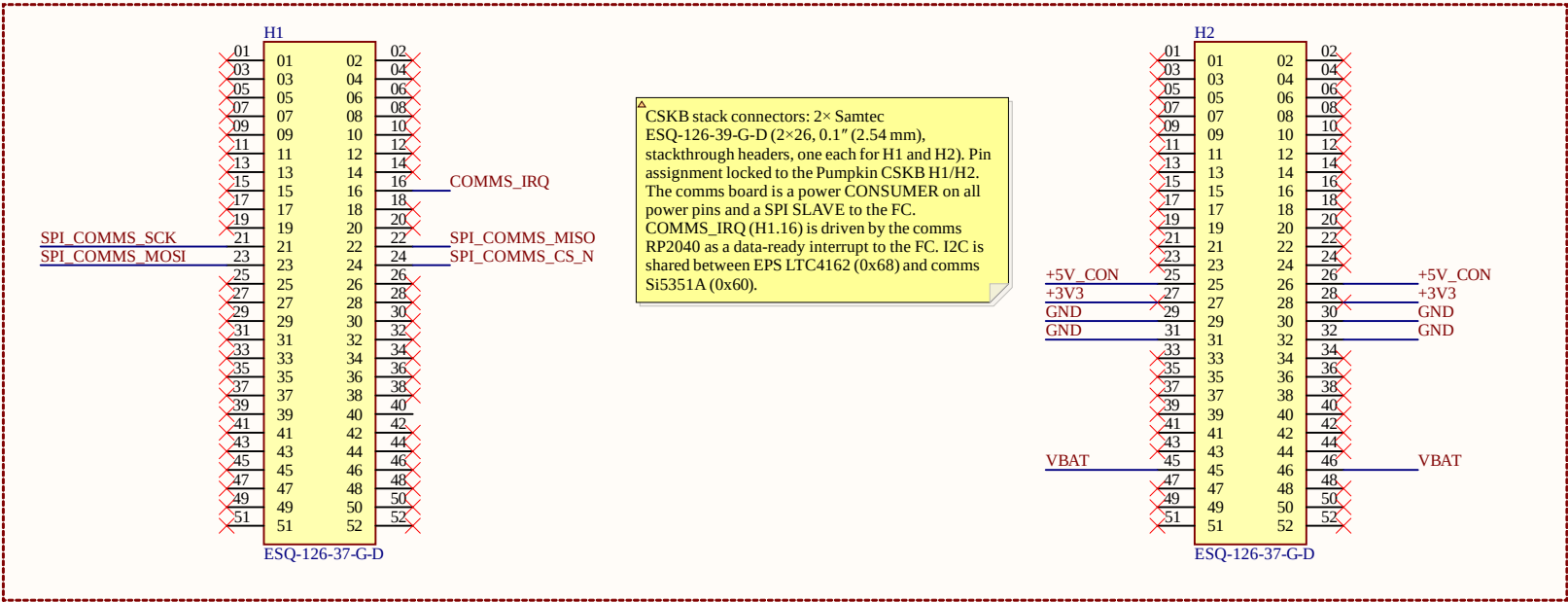


Title			CubeSat Comms -- Digital Control	
Size	Number		Revision	
A3			A	
Date:	8/15/2026		Sheet5 of 7	
File:	Digital Control.SchDoc		Drawn By: Nick Grabbs - KI5YNG	

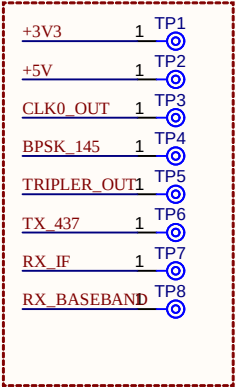
Connectors

CONNECTORS — CSKB H1 + H2 stack connectors for inter-board power and signal distribution, RF connectors for TX and RX antenna ports, and test points for bench validation. All power (+3V3, +5V) arrives on H2; I2C, SPI, and COMMS_IRQ route through H1. The comms board is a power CONSUMER — it receives regulated rails from the EPS.

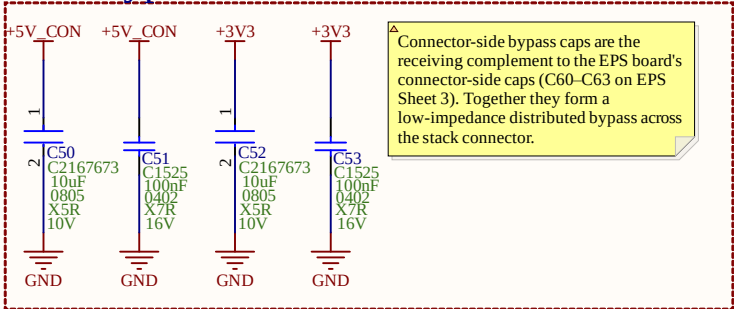
CSKB H1 + H2 Stack Connectors



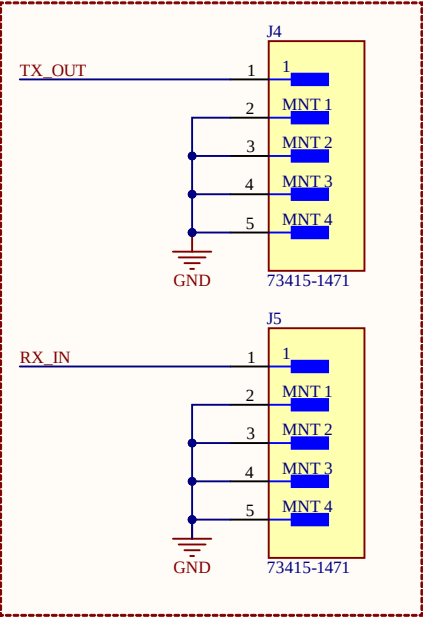
Test Points



Bulk Bypass at Stack Connector



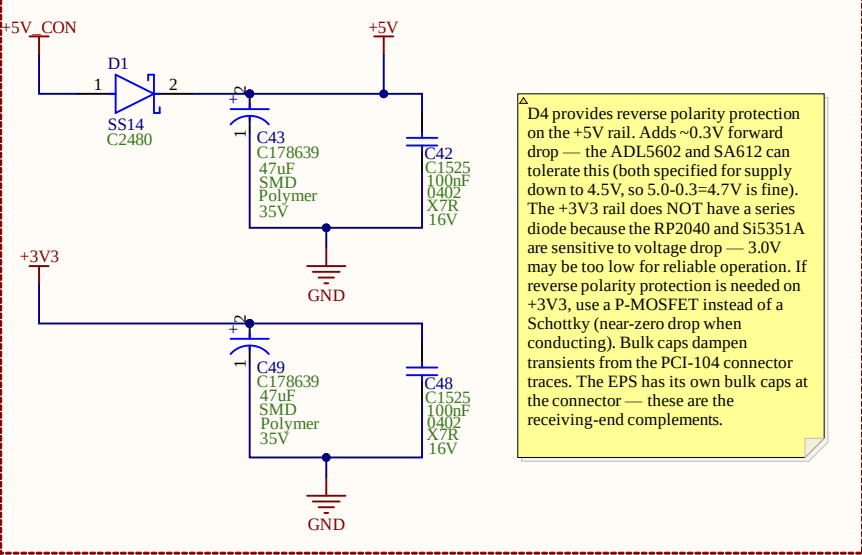
RF Connectors



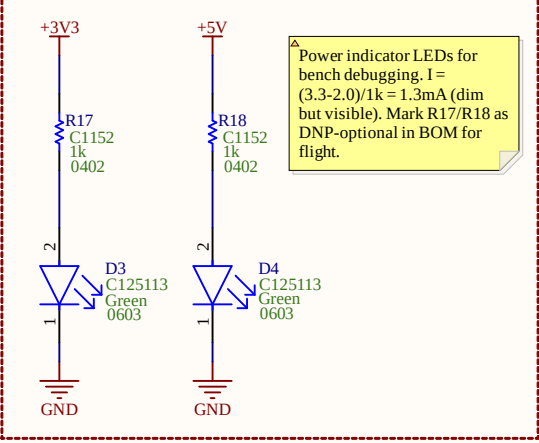
POWER DISTRIBUTION

POWER DISTRIBUTION — The comms board receives +3V3 and +5V from the EPS via the CSKB stack connectors. No on-board voltage regulation. This sheet provides input protection, bulk bypass capacitors, and distributes power to all subsystems. Power budget: ~150 mA on +3V3 (RP2040 + Si5351A + logic), ~140 mA on +5V (PSA4-5043+ LNA + ADL5602 MMIC + tripler bias). Total: <300 mA — well within EPS TPS62933F capacity (3A per rail).

Input Protection and Filtering



Power Indicator LEDs



Pico Module Power Feed

